Taizoon Aliasgar Chunawala

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Education

Virginia Tech, Blacksburg, VA — Aug 2021 – Expected May 2027
Ph.D., Mechanical Engineering
Dissertation: Structurally Augmented Optimal and Safety-Critical Control for Robotic Systems Under Dynamic and Environmental Uncertainties

University of Southern California, Los Angeles, CA — Dec 2019
M.S., Aerospace & Mechanical Engineering (Dynamics & Control)

BITS Pilani, Pilani, India — Jun 2016
B.E. (Hons.), Mechanical Engineering

Technical Skills

Control & Planning: Nonlinear/Robust/Distributed MPC, Trajectory Optimization (Direct Collocation), Contact Planning, Whole-Body Control (QP), Optimal Control
Estimation: Kalman Filtering, Sensor Fusion (Kinematics + IMU, Vision + IMU), State Estimation
Programming: C++ (real-time, multithreaded), Python, MATLAB/Simulink
Solvers & Libraries: CasADi, IPOPT, OSQP, qpOASES, Pinocchio, Eigen
Frameworks & Tools: ROS/ROS2, MuJoCo, Raisim, OpenCV, Linux, Git, CMake
Hardware: Unitree Go2, Unitree A1, Kinova arm
Modeling & Design: Rigid-Body Dynamic Modeling, URDF, SolidWorks

Research Experience

Graduate Research Assistant — Feb 2022 – Present
Agile Robots for Control, Optimization & Learning Lab, Virginia Tech, Blacksburg, VA

- **Safety-Critical Distributed NMPC for Quadrupedal Cart Pulling** *(dissertation, ongoing)*
  - Developing a **safety-critical distributed NMPC** planner for a quadruped pulling a cart through **obstacle-filled environments**, with a **modular multi-agent formulation** for actuated or unactuated agents.
  - *Planned:* **adaptive** identification of cart loading with online adjustment of control parameters.
- **Multi-Rate NMPC for Wall-Supported Bipedal Locomotion of a Quadruped** *(lead author)*
  - Developed a layered multi-rate NMPC that **jointly plans footstep locations and single-rigid-body (SRB) trajectories**, achieving **2.9× higher success** than a Raibert-heuristic baseline across **250 randomized rough-terrain trials** in simulation (Unitree A1).
  - Solved the NMPC in **8.3 ms at 100 Hz** (CasADi/IPOPT), tracked by a **1 kHz** QP-based whole-body controller enforcing full-order dynamics.
  - Built a multithreaded real-time C++ stack and 1 kHz kinematics–IMU state estimator for **preliminary hardware deployment on a Unitree Go2**: crouch to wall-supported bipedal walking, balanced for 60+ s. [paper] [video][code]
- **Robust MPC for a Tremor-Suppression Exoskeleton** *(lead author)*
  - Designed a robust MPC that plans across the **worst-case tremor disturbances** of a convex uncertainty set built from tremor recorded in essential-tremor patients.
  - Introduced **state-dependent cost weighting** that prioritizes the disturbance driving tracking error, cutting RMSE **20–33%** vs. observer-based MPC and **up to 47%** on patient-derived tremor (simulation).
  - Reduced the 9-DoF arm–exoskeleton model to 2 wrist coordinates for real-time control (14.6 ms average solve per 20 ms step); stress-tested under sensor noise and 50% actuation latency. [video][code]

- **Decomposition-Based NMPC for Quadrupedal Loco-Manipulation** *(co-author)*
  - Implemented the real-time interface between a **60 Hz decomposition NMPC** planner and a **500 Hz QP-based WBC**, passing optimal ground reaction forces and interaction wrenches for tracking.
  - Supported hardware validation on a Unitree Go2 carrying a 4.4 kg Kinova arm under unmodeled terrain and 125% payload uncertainty. [video]
- **Safety-Critical Distributed NMPC for Multi-Robot Quadrupedal Teams** *(co-author)*
  - Co-developed a **100 Hz distributed nonlinear MPC** with safety constraints for Unitree A1 robot teams.
  - Diagnosed and fixed real-time data pipeline and one-step-delay communication issues across multi-threaded processes, enabling stable hardware experiments under latencies up to 6 ms. [video]
- **Adaptive MPC for Payload Transportation by Quadrupedal Robots** *(co-author)*
  - Designed payload-transportation experiments and ran additional hardware experiments on the Unitree A1 to address reviewer comments (payloads up to 109% of robot mass). [video]

Research Assistant Feb 2020 – Dec 2020
Dynamic Robotics & Control Lab, University of Southern California, Los Angeles, CA

- Formulated and solved **jumping trajectory optimization via direct collocation** for an 18-DOF quadruped on compliant terrain, achieving target jump heights and stance-to-flight transitions in simulation.

Research Assistant May 2018 – May 2019
Autonomous Microrobotic Systems Lab, University of Southern California, Los Angeles, CA

- Developed adaptive and predictive controllers for piezo-actuated flapping-wing microrobots, implemented in Simulink Real-Time for closed-loop flight experiments.

Professional Experience

Control Engineering Intern Jun 2019 – Aug 2019
Polybee, Singapore

- Developed a vision-based localization system with **millimeter-level accuracy** for micro-quadrotors performing autonomous pollination in vertical farms.
- Fused vision-based localization with IMU data for robust state estimation and smooth trajectory tracking in confined indoor spaces.

Operations Manager Jul 2016 – Nov 2017
Tata Steel Global Wires, Tarapur, India

- Optimized production scheduling across wire grades and gauges, improving line uptime and throughput.

Selected Publications

T. Chunawala et al., "Multi-Rate Nonlinear Model Predictive Control for Wall-Supported Bipedal Locomotion of Quadrupedal Robots," *IEEE/RSJ IROS*, 2026. [arXiv]

T. Chunawala et al., "Robust Model Predictive Control of a Tremor-Suppression Exoskeleton under Tremor-Induced Disturbances," *ASME Journal of Dynamic Systems, Measurement, and Control*, accepted, 2026.

R. Sambhus, K. K. Mehta, A. M. Sadeghi, B. M. Imran, J. Kim, **T. Chunawala** et al., "A Nonlinear MPC Framework for Loco-Manipulation of Quadrupedal Robots With Non-Negligible Manipulator Dynamics," *IEEE Robotics and Automation Letters*, 2026.

B. M. Imran, J. Kim, **T. Chunawala** et al., "Safety-Critical and Distributed Nonlinear Predictive Controllers for Teams of Quadrupedal Robots," *IEEE Robotics and Automation Letters*, 2025.

L. Amanzadeh, **T. Chunawala** et al., "Predictive Control with Indirect Adaptive Laws for Payload Transportation by Quadrupedal Robots," *IEEE Robotics and Automation Letters*, 2024.

Honors

Pratt Fellowship, Department of Mechanical Engineering, Virginia Tech 2021–22